Koneru Lakshmaiah Education Foundation
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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

Experiment -2

AIM: 2 a)Enabling Multi-Factor Authentication to Secure Your Access
b) Creating First Linux Instance
c) S3 Bucket Creation and Upload Objects
I. Create an S3 Bucket for file storage.
II. Upload 5 objects with different file extensions.

Description:

Multi-Factor Authentication (MFA) provides an additional layer of security to an AWS account. It requires users to enter a verification code along with their password, preventing unauthorized access.
Amazon EC2 (Elastic Compute Cloud) allows users to create and manage virtual servers in the cloud. A Linux instance provides a virtual Linux operating system for running applications and executing commands.
Amazon S3 (Simple Storage Service) is a cloud storage service used to store and retrieve files. Files stored in S3 are called objects, and they are organized inside buckets.

Steps:

a) Enabling Multi-Factor Authentication (MFA)

Steps:

1. Log in to AWS Console.
2. Open IAM → Users.
3. Select the IAM user.
4. Go to Security credentials.
5. Click Assign MFA device.
6. Select Authenticator app.
7. Scan the QR code.
8. Enter MFA codes and click Add MFA.

b) Creating First Linux Instance

Steps:

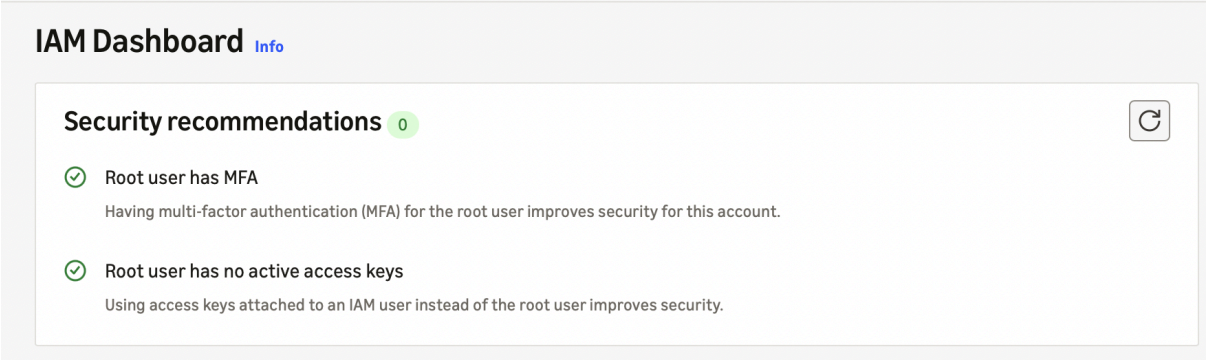
1. Log in to AWS Console.
2. Open EC2 → Launch instance.
3. Enter the instance name.
4. Select Amazon Linux.
5. Choose a Free Tier eligible instance type.
6. Create a key pair.
7. Configure security settings.
8. Click Launch instance.
9. Click Connect to access Linux.

c) S3 Bucket Creation and Upload Objects

Steps:

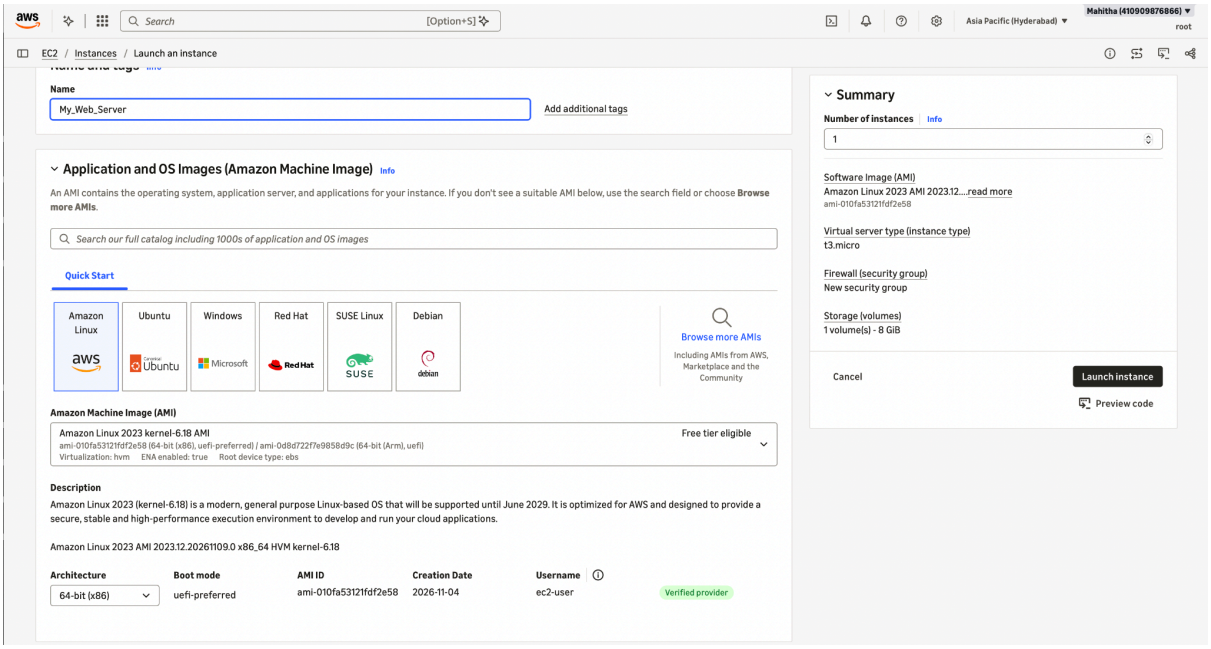
1. Open AWS → S3.
2. Click Create bucket.
3. Enter a unique bucket name.
4. Keep default settings and click Create bucket.
5. Open the created bucket.
6. Click Upload → Add files.
7. Select five files (.txt, .jpg, .pdf, .csv, .py).
8. Click Upload.
9. Verify the uploaded objects.

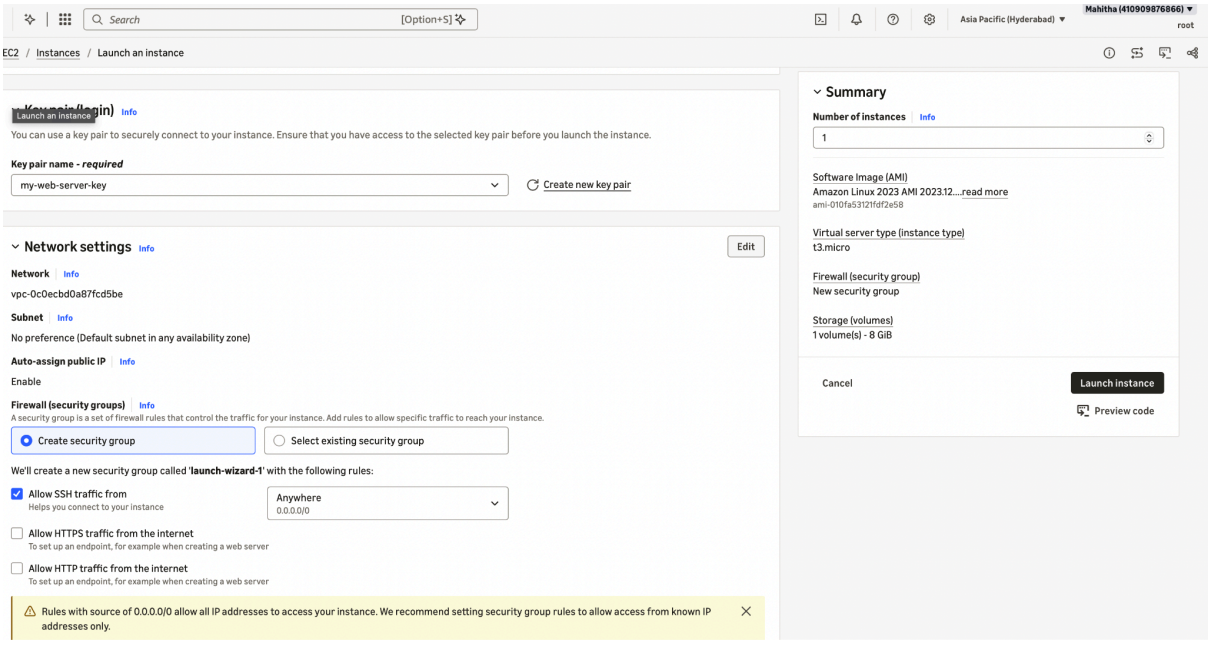
Output:
2a)

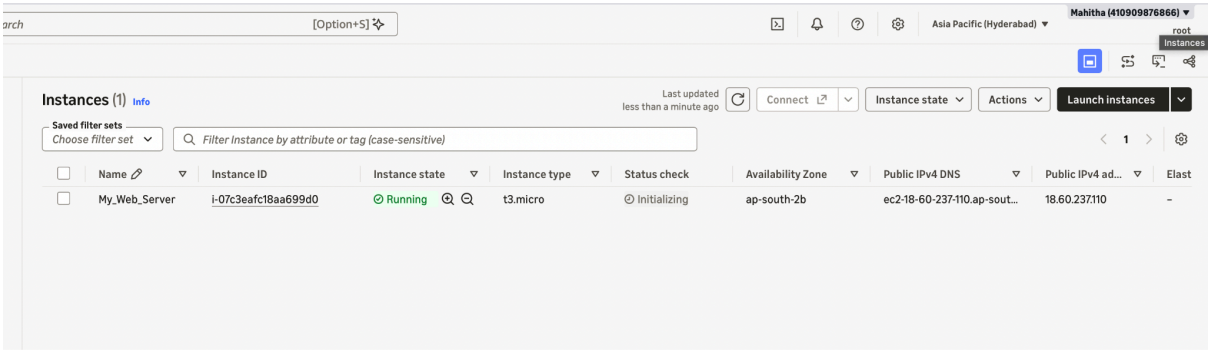


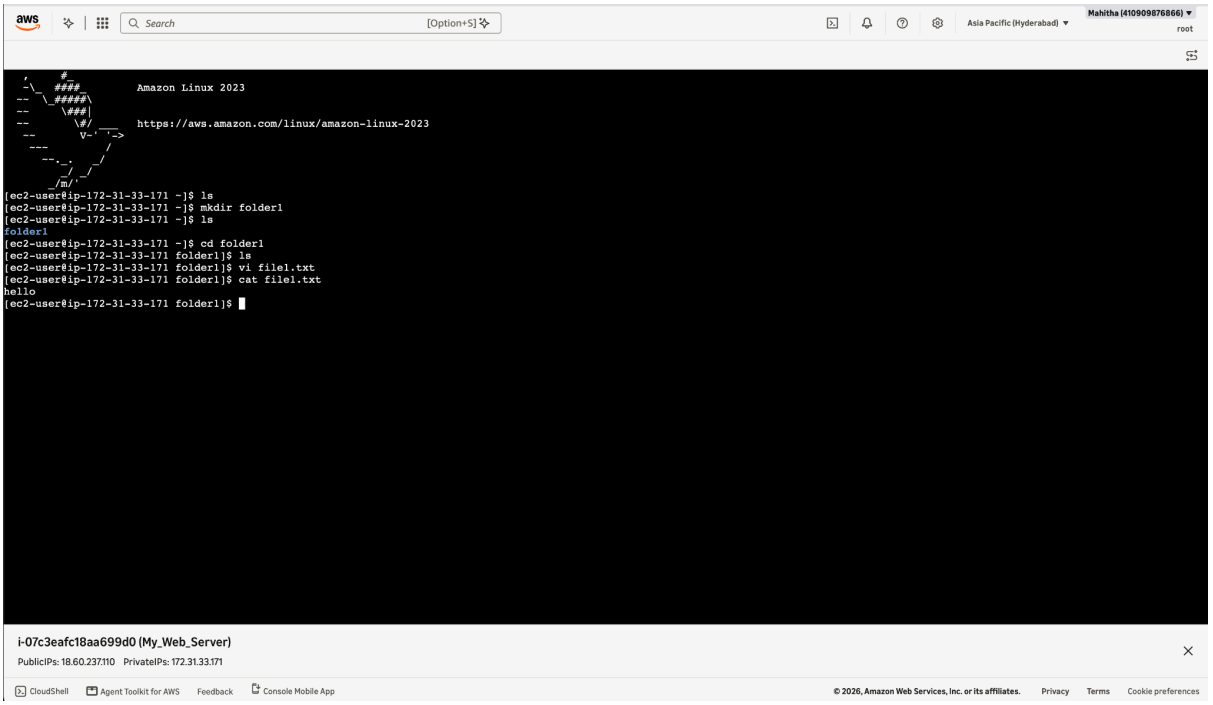
MFA established

2b)


aws
Search
[Option+S]
Asia Pacific (Hyderabad)
Mahitha (410909876866)
root
EC2 / Instances / Launch an instance
Name
My_Web_Server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
ami-010fa53121fdf2e58 (64-bit (x86), uefi-preferred) / ami-0d8d722f7e9858d9c (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Free tier eligible
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20261109.0 x86_64 HVM kernel-6.18
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-010fa53121fdf2e58
Creation Date
2026-11-04
Username
ec2-user
Verified provider
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-010fa53121fdf2e58
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

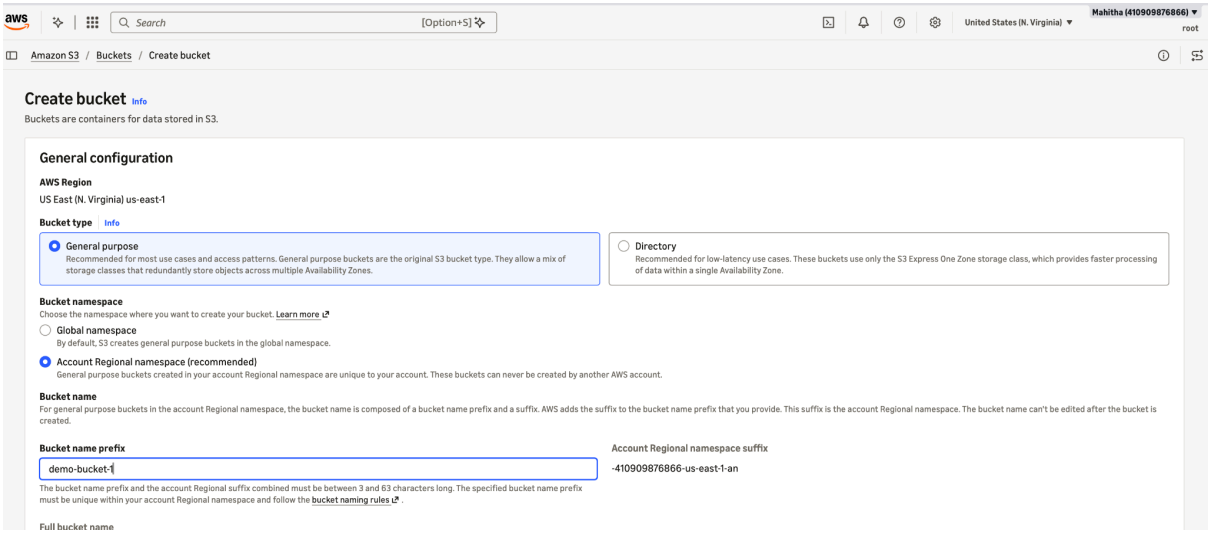

Search
[Option+S]
Asia Pacific (Hyderabad)
Mahitha (410909876866)
root
EC2 / Instances / Launch an instance
Launch an instance
Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
my-web-server-key
Create new key pair
Network settings Info
Edit
Network Info
vpc-0c0ecbd0a87fcd5be
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-1' with the following rules:
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-010fa53121fdf2e58
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code



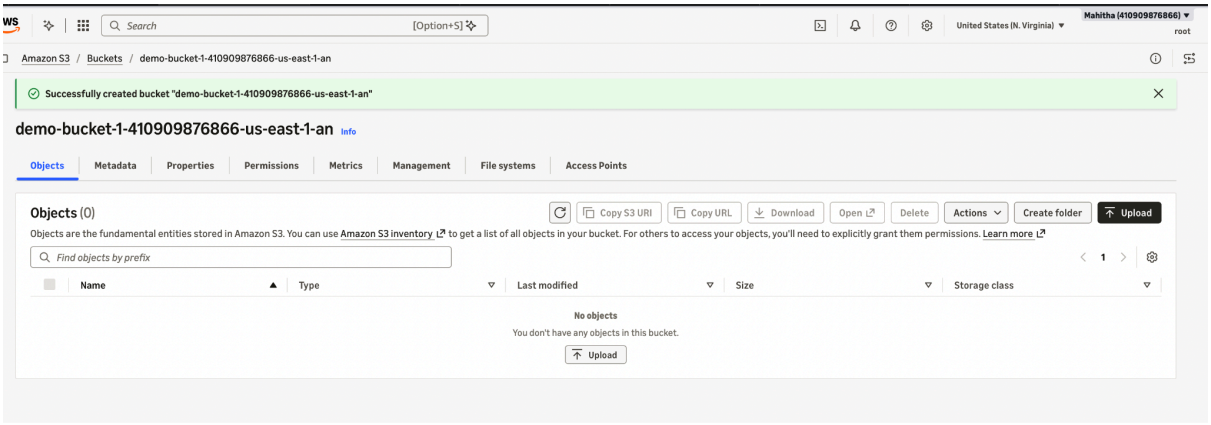


Creating instance, entering key pair name and connecting to instance

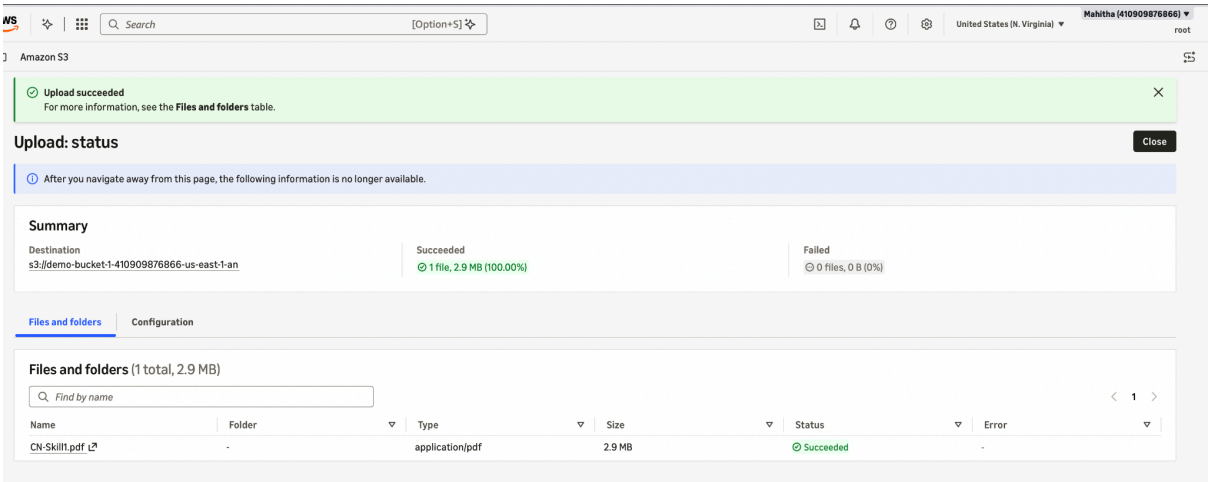
2c)



Naming bucket



Creating bucket



Uploading files